

EEET202

CHINES & TRANSFORMERS

MODULE:2

Lecture:8 – Commutation

Performance characteristics of DC generators

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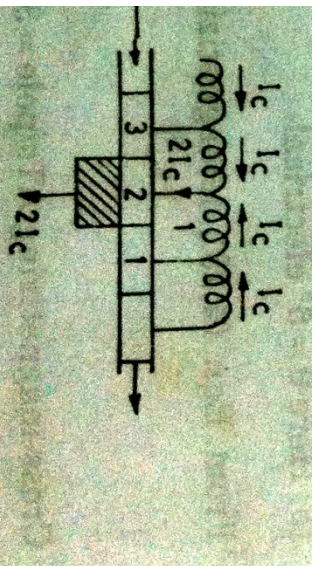
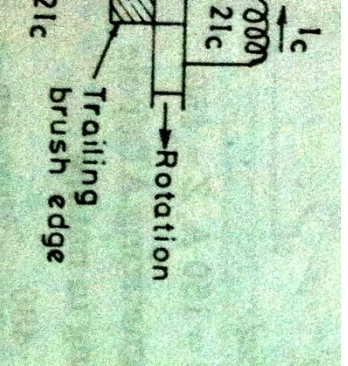
Commutation

ature conductors are alternately comes under , direction of current changes.

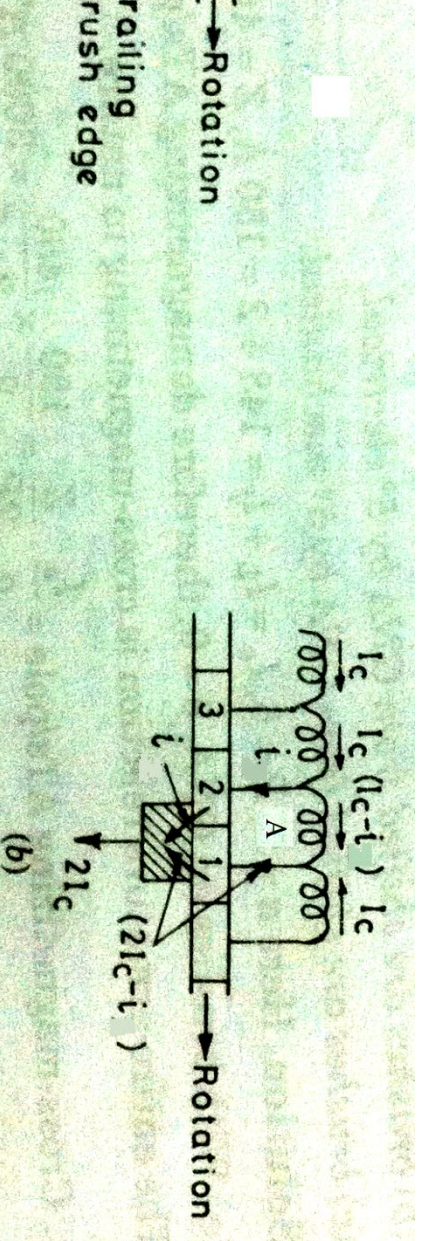
circuit, we get unidirectional current.

of converting bidirectional current in the uctors to unidirectional current in the external ins of commutator segments and brushes is **ation process**.

, the reversal of current in the armature coil by mmutator segments and brushes is known as

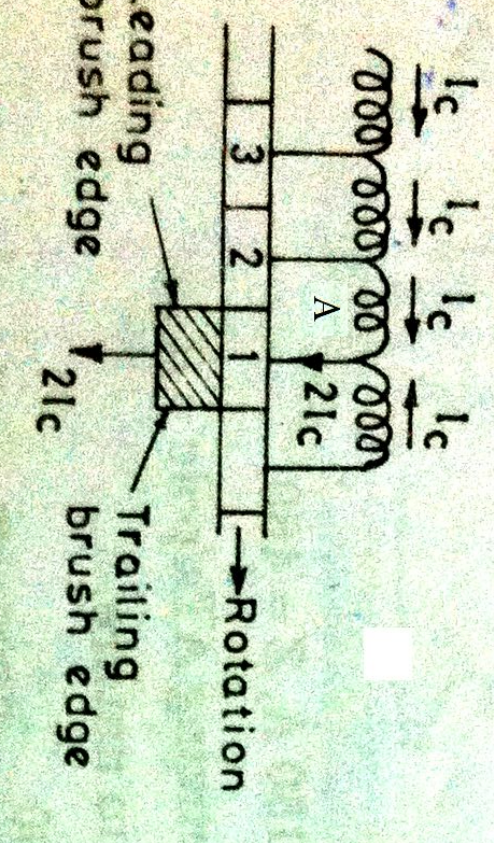


Stages of commutation process



4. Pertaining to the illustration of commutation process.

ation of Different Stages

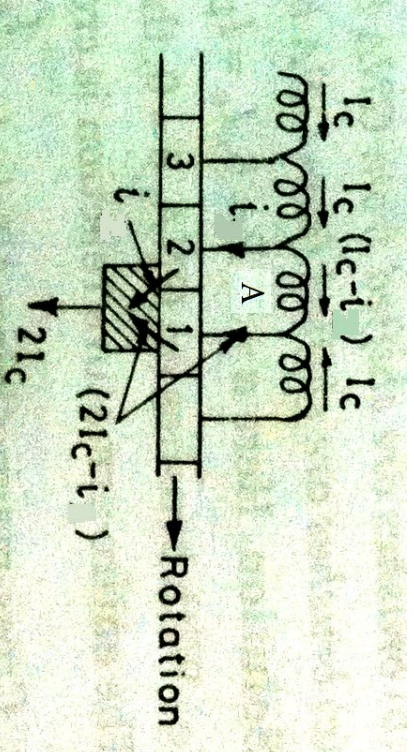


on commutator segment 1.

ducts a current of $2I_c$ to the brush.

and I_c from the adjacent coil.

on of Different Stages (contd..)



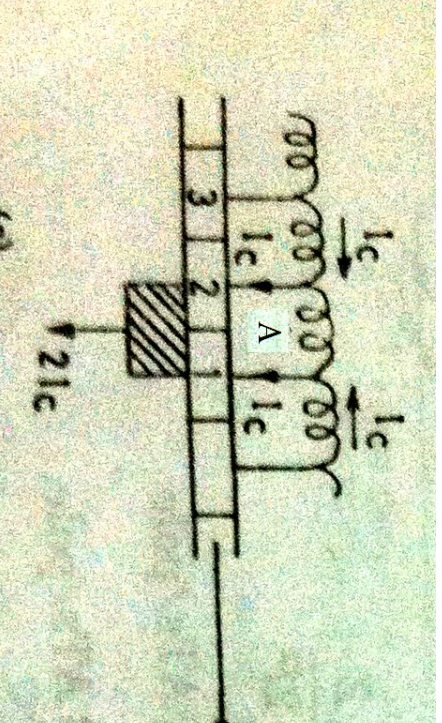
rotates, brush makes contact with segment
ort circuits coil A.

ne-fourth on segment 2 and three-fourth

ts a current of $2I_c$; i through segment 2 and
h segment 1.

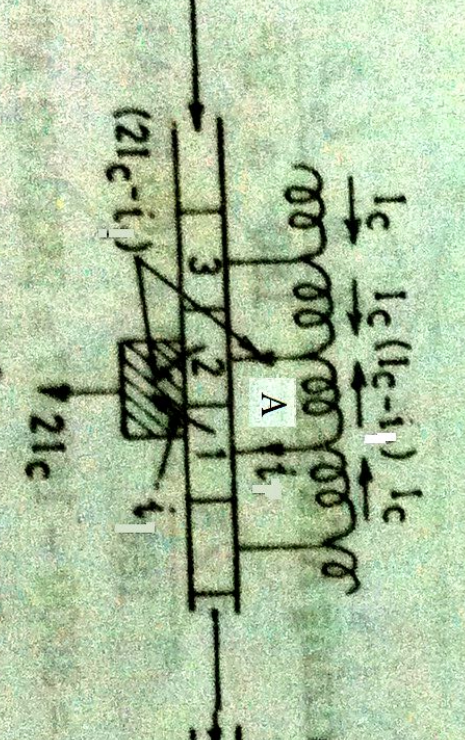
A is reduced to $(I_c - i)$ from I_c .

on of Different Stages (contd..)



one-half on segment 1 and one-half on
ts a current of $2I_c$; I_c through segment 1
h segment 2.
A is reduced zero.

tion of Different Stages (contd..)

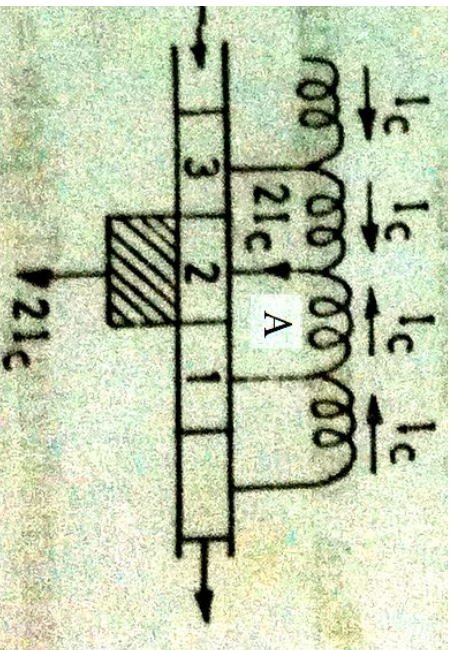


one-fourth on segment 1 and three-fourth

ts a current of $2I_c$; i through segment 1 and
h segment 2.

a current of $(I_c - i)$ in the opposite direction.

tion of Different Stages (contd..)



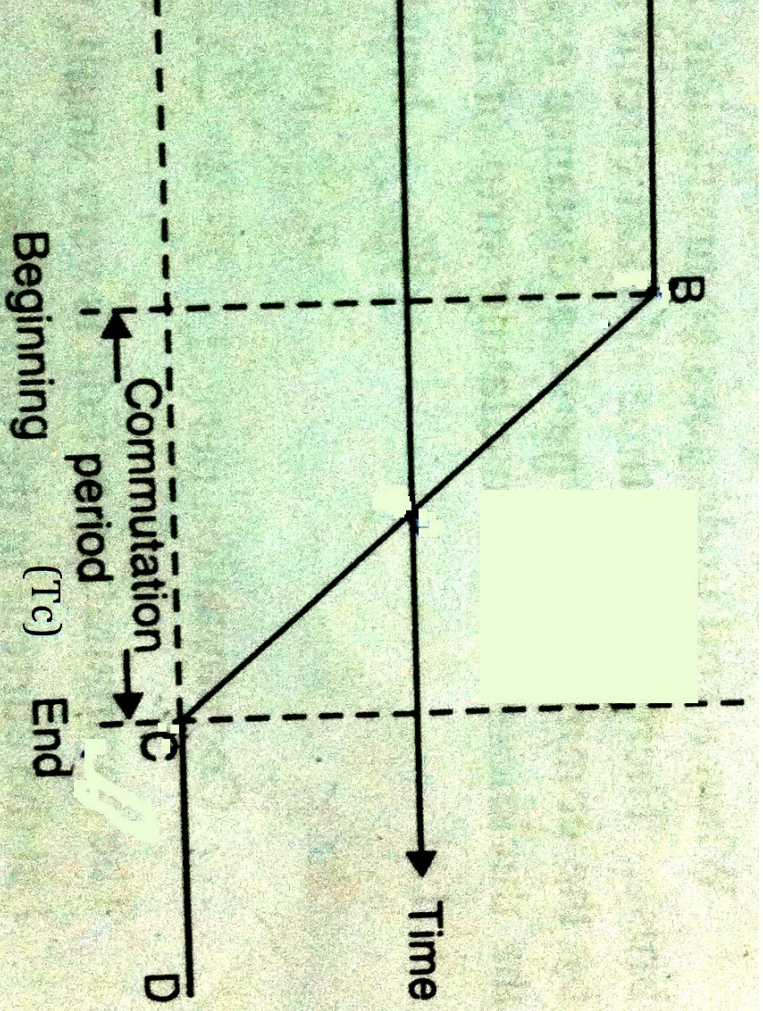
lly on segment 2.

s a current of $2I_c$; I_c through coil A and I_c coil.

current of I_c but in the reverse direction.

undergone commutation.

undergo commutation in this way as it passes



time curve of coil undergoing commutation

Delayed Commutation

Inducted coil current changes from $+I_c$ to $-I_c$ in period (T_c).

Current change in small interval of time (T_c), induced in the coil due to its inductance (L_c).

Induced EMF is called **reactance voltage**.

It is given by,

$$e_c = L_c \frac{di_c}{dt}$$

ed Commutation (contd..)

z's Law, the effect opposes the cause.

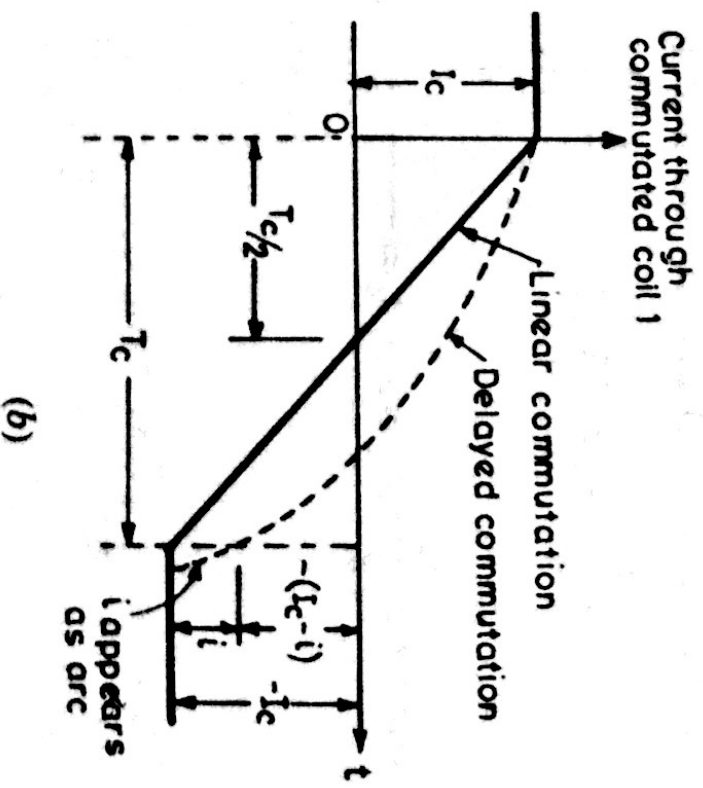
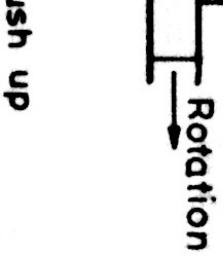
is reactance voltage and cause is reversal of voltage opposes the reversal of current in the .
sal of current is delayed.

mutation is called **delayed commutation**.

ent in the commutated coil will not reach $-I_c$ with

ment-1 carries some current during the breaking with segment-1.

arking at the brushes and overheating of sh contacts.



s. Time curve of Delayed Commutation

Methods to Improve Commutation

Interpoles: The effect of reactance voltage is reduced by increasing the contact resistance between the brushes and the commutator. This is achieved by using carbon

Interpoles: In this method, an arrangement is made to produce a reverse reactance voltage by producing a reverse emf in the coil undergoing commutation. The reversing emf is produced by the use of **interpoles**. As their polarity is opposite to the main poles, they induce an emf opposite to the reactance voltage, thus opposing the reactance voltage. As the main pole ahead, they induce an emf opposite to the reactance voltage, thus opposing the reactance voltage. This is achieved by using carbon

Applications of DC Generators

Excited Generators

Wide output voltage range, it is suitable for the control of DC motors

Generators

Terminal voltage is almost constant independent of load, it can be used for **Battery charging**

Field of AC generator

Generators

Used to compensate voltage drop in feeders.